Second Calls and Mutual Aid

```
# A tibble: 11 x 2
   `Capability of Unit (eResponse.07)`     n
   <chr>                                <int>
 1 Ambulance (ALS)                       5036
 2 Ambulance (BLS)                        828
 3 Ambulance (Critical Care)                1
 4 Call Turn Over / Mutual Aid            147
 5 Critical Care Ambulance                  2
 6 First Response                          25
 7 First Response (ALS)                   169
 8 First Response (BLS)                    25
 9 Intercept                                2
10 Intercept (ALS)                          4
11 <NA>                                     7
```

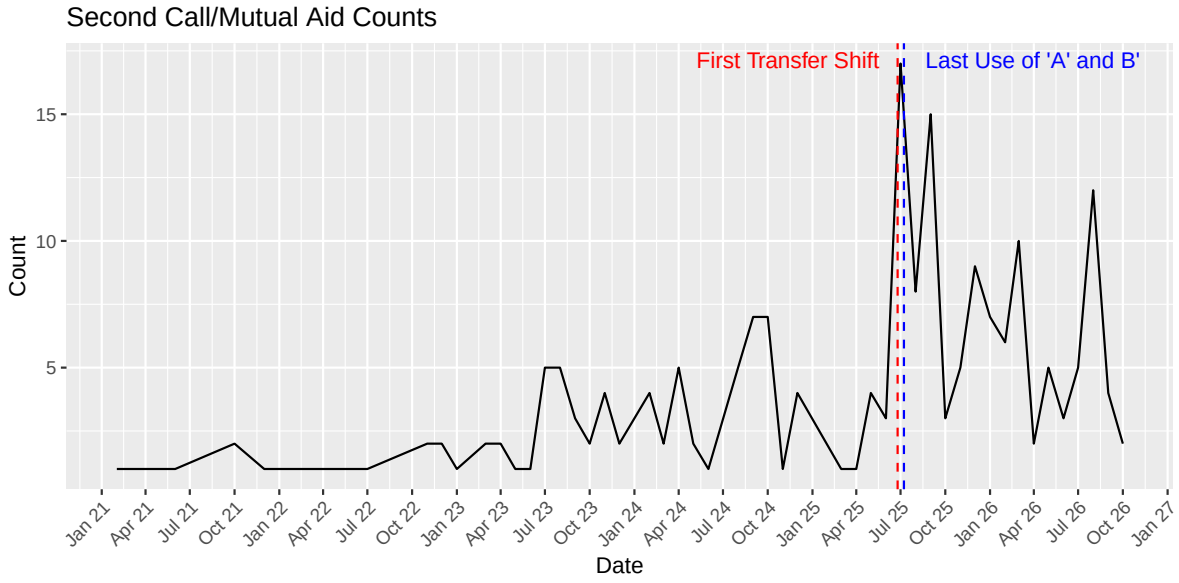
```
# A tibble: 9 x 2
  `EMS Shift (itResponse.005)`                n
  <chr>                                   <int>
1 911 Day                                   821
2 911 Night                                 426
3 A                                        2516
4 B                                        1545
5 Event/Standby                              29
6 No Shift (Second Call)                      6
7 No Shift (Second Call/Call-In Transfer)    49
8 Transfer Shift                            262
9 <NA>                                      592
```

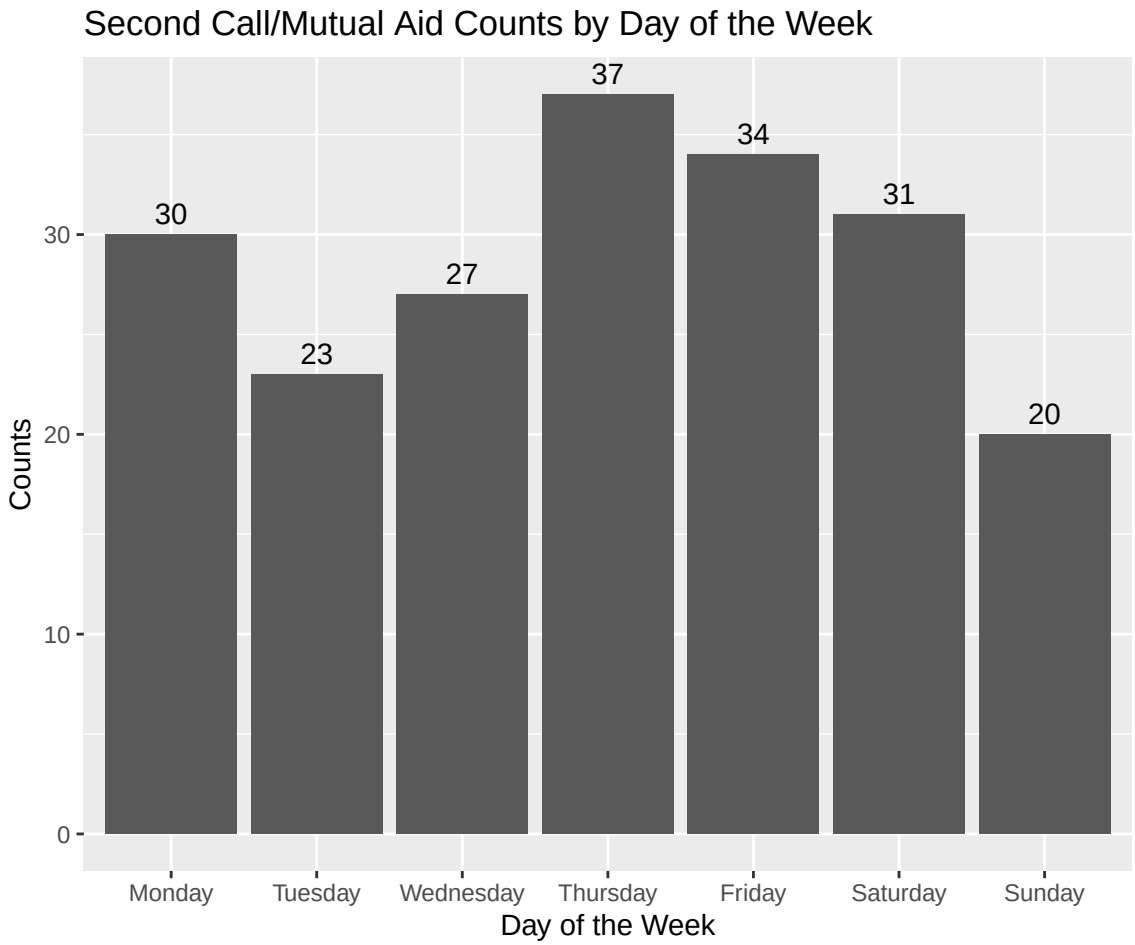
The total number of incidents not taken by the scheduled EMS crew from 2021-Present is 153.

```
# A tibble: 4,061 x 3
   `Incident Date` `EMS Shift (itResponse.005)` `Type of Call (eResponse.05)`
   <date>          <chr>                        <chr>
 1 2021-01-01      B                            911 Response (Scene)
 2 2021-01-03      B                            911 Response (Scene)
 3 2021-01-04      B                            911 Response (Scene)
 4 2021-01-04      A                            911 Response (Scene)
 5 2021-01-04      B                            911 Response (Scene)
 6 2021-01-05      B                            911 Response (Scene)
 7 2021-01-05      A                            911 Response (Scene)
 8 2021-01-05      A                            911 Response (Scene)
 9 2021-01-06      A                            911 Response (Scene)
10 2021-01-06      A                            911 Response (Scene)
# i 4,051 more rows
```

```
[1] 10
```

We were wondering what EMS Shift "A" and "B" were referring to. The last use of them was 2025-07-08.

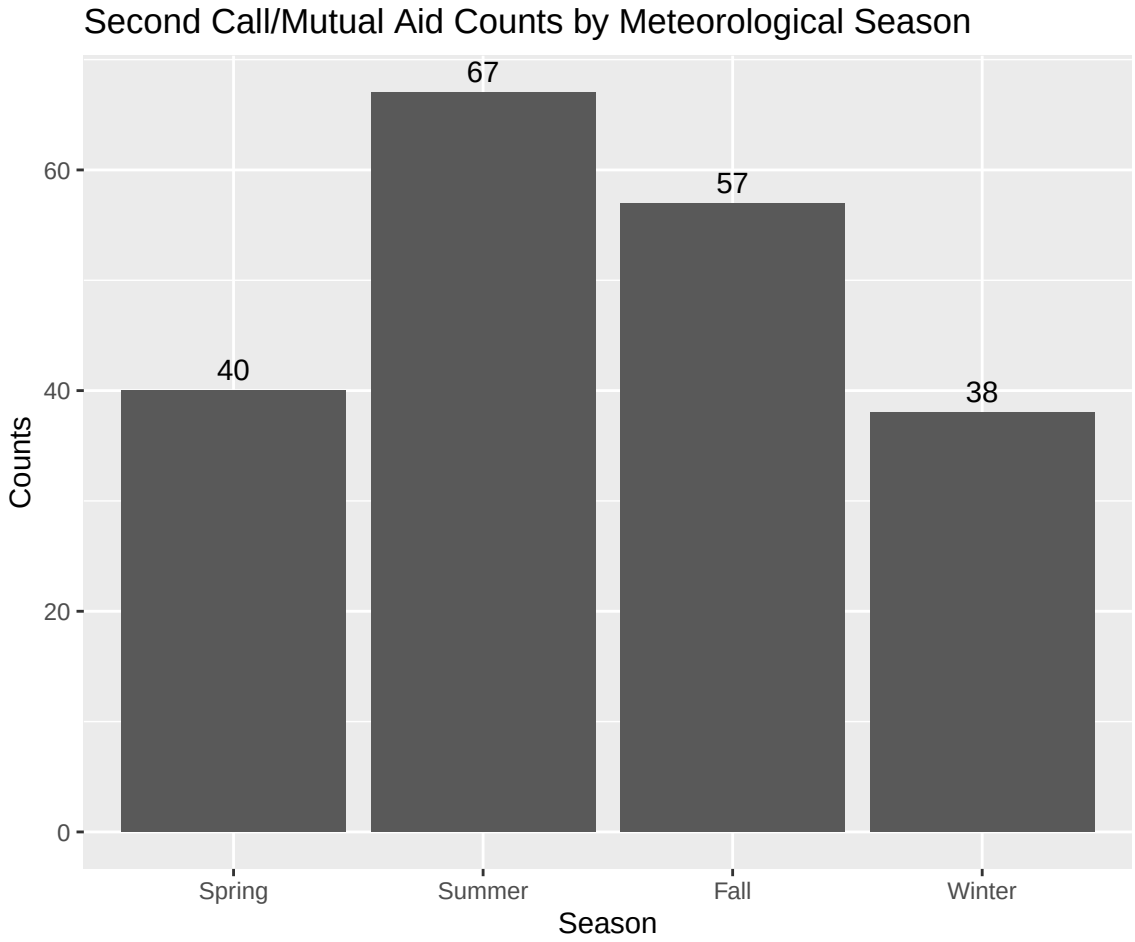





```
    Chi-squared test for given probabilities

data:  table(second_aid_filter$`Incident Day Name`)
X-squared = 7.4455, df = 6, p-value = 0.2816
```

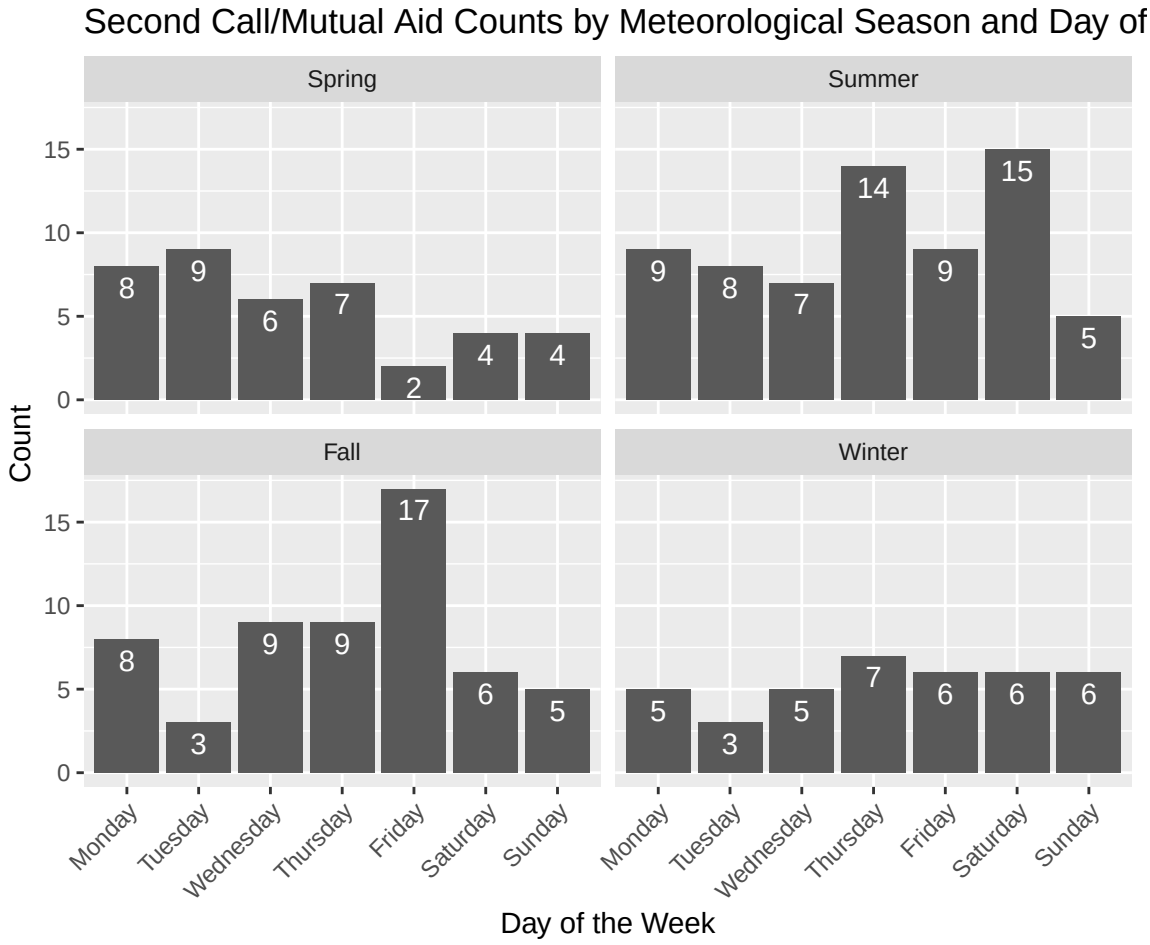
No clear/obvious trend here. We ran a Chi-Squared test and found that there is no significant difference between the amount of calls on any day.




```
    Chi-squared test for given probabilities

data:  table(second_aid_filter$Season)
X-squared = 11.505, df = 3, p-value = 0.009287
```

Here there does appear to be a difference between the seasons and a Chi-Squared test showed that there was a significant difference between the amount of calls in a season.



Spring

```
    Chi-squared test for given probabilities

data:  table(spring$`Incident Day Name`)
X-squared = 6.55, df = 6, p-value = 0.3645
```

Summer

```
    Chi-squared test for given probabilities

data:  table(summer$`Incident Day Name`)
X-squared = 8.3284, df = 6, p-value = 0.215
```

Fall

```
    Chi-squared test for given probabilities

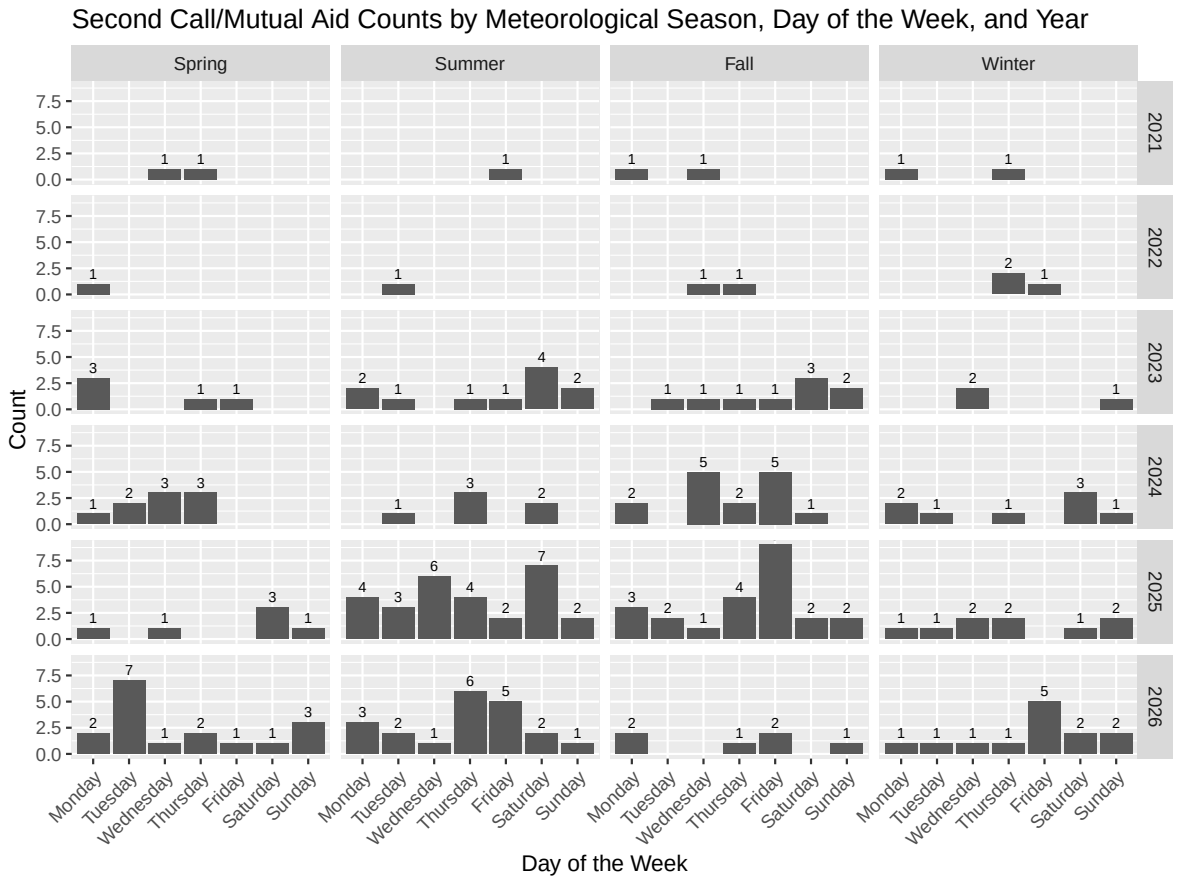
data:  table(fall$`Incident Day Name`)
X-squared = 14.842, df = 6, p-value = 0.02152
```

Winter

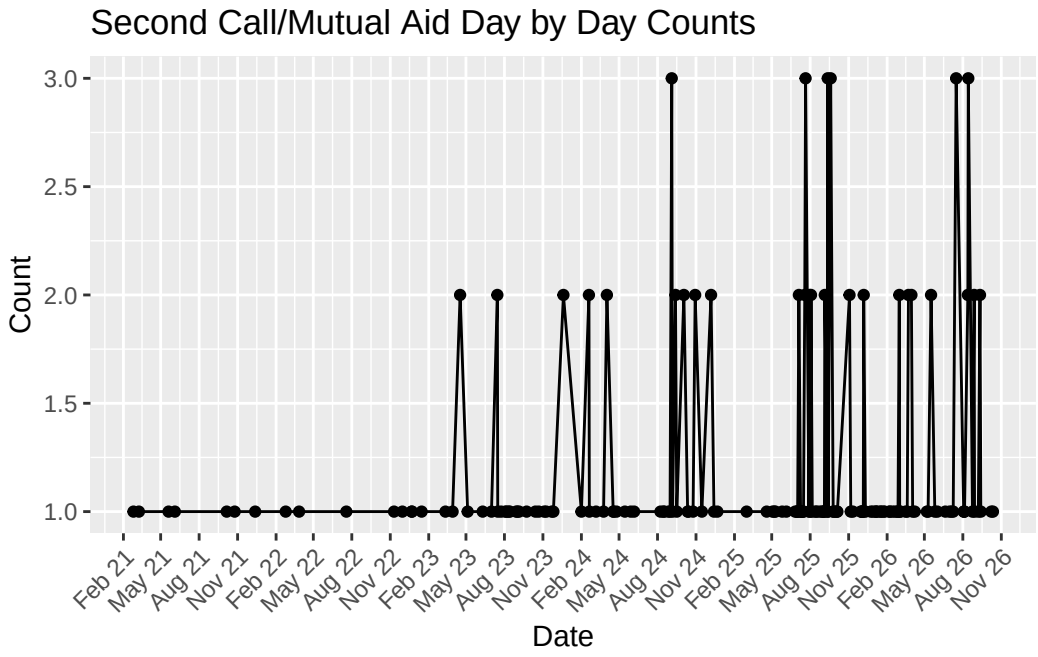
```
    Chi-squared test for given probabilities

data:  table(winter$`Incident Day Name`)
X-squared = 1.7895, df = 6, p-value = 0.938
```

This does show that there is a significant difference for the day of the week in the fall.



However, it could possibly be explained by the the number of Friday calls in the fall of 2025.




```
# A tibble: 6 x 2
  period         n
  <date>     <int>
1 2024-09-04     3
2 2025-07-21     3
3 2025-09-12     3
4 2025-09-19     3
5 2026-07-16     3
6 2026-08-14     3
```

The last thing we looked at here was the number of calls on a given day. We can see that the max was 3 in one day.

Next steps would be to look into these days and see if we can determine any reasons as to why (weather, time of year, etc…). We have also discussed using modeling techniques to determine if/when you should schedule a second crew (output could be a % chance that you will need the second crew).